MIL-S-51270
4 April 1968

MILITARY SPECIFICATION

STANNIC OXIDE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers reagent grade stannic oxide.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-C-96 - Cans, Metal, 28 Gage and Lighter.
PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-T-60 - Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.
MIL-S-50005 - Stannic Oxide, Technical.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Chemical characteristics. Stannic oxide shall conform to the chemical characterisitics of table I when tested as specified therein.

Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as SnO_2)	99.0	----	4.2.4.1
Water-soluble salts	----	0.50	4.2.4.2
Free alkali (as NaOH)	----	0.20	4.2.4.3
Chloride (Cl)	----	0.005	4.2.4.4
Sulfate (SO_4)	----	0.030	4.2.4.5

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of section 5 for which specific inspection has not been provided in the specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the stannic oxide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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4.2.2 Sampling for test. Sampling shall be conducted in accordance with table II. Approximately equal portions of stannic oxide shall be taken from each container in the sample, composited, and placed in a suitable clean, dry container labeled to identify the lot and containers from which it was taken. The total composited sample shall be no less than 1/4 pound.

Table II. Sampling for test

Number of containers in lot or batch	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
over 20,000	20

4.2.3 Inspection procedure. The composited sample of stannic oxide shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted follows:

4.2.4.1 Assay (as SnO_2). Determine percent stannic oxide in accordance with the assay procedure specified in MIL-S-50005.

4.2.4.2 Water-soluble salts. Weigh to the nearest 10 milligrams (mg) approximately 1 gram (g) of the specimen. Add 20 milliliters (ml) of concentrated ammonium hydroxide and 50 ml of water. Cover with a watch glass and digest on a steam bath for 1/2 hour. Cool, filter, and wash with water to make 100 ml of filtrate. Evaporate a 50-ml aliquot of the filtrate to dryness in a tared crucible at 600° ± 10° C, cool in a desiccator, and weigh. Calculate the percent water-soluble salts as follows:

$$\text{Percent water-soluble salts} = \frac{200\ (A-B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

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4.2.4.3 Free alkali (as NaOH). Weigh to the nearest 0.1 g approximately 5 g of the specimen. Add 80 ml of water and 20 ml of 0.1N hydrochloric acid and digest for 5 minutes at 50° ± 2° C with frequent shaking. Titrate the excess acid with 0.1N sodium hydroxide solution using phenolphthalein indicator. Calculate the percent free alkali as follows:

$$\text{Percent free alkali} = \frac{4\ (AB-CD)}{W}$$

where: A = Milliliters of hydrochloric acid added,
B = Normality of hydrochloric acid,
C = Milliliters of sodium hydroxide solution used,
D = Normality of sodium hydroxide solution, and
W = Weight of specimen in grams.

4.2.4.4 Chloride. Weigh 0.20 g of the specimen and mix with 20 ml of water. Filter through a filter crucible and to the clear filtrate (centrifuge if still not clear), add 1 ml of concentrated nitric acid and 1 ml of 1.7-percent silver nitrate solution. Any turbidity produced shall be no greater than that produced by a control containing 0.01 mg of Cl (chloride) and the same quantities of reagents in the same final volume as in the specimen test.

4.2.4.5 Sulfate. Weigh 1.00 g of the specimen and mix with 20 ml of water. Filter through a filter crucible (centrifuge if filtrate is not clear) and dilute the filtrate to 50 ml with water. To a 10-ml aliquot of the solution, add 1 ml of 1N hydrochloric acid and 1 ml of 12-percent barium chloride solution. The turbidity after 10 minutes shall be no greater than that produced by a control containing 0.06 mg of sulfate (SO_4) ion and the same quantities of reagents in the same final volume as in the specimen test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One pound ± 0.05 ounce of stannic oxide shall be packaged in either group A, class 1, type e, style 1, grade optional glass bottles or in group A, class 2, style 1, grade optional plastic bottles of PPP-C-186. Closure shall be in accordance with closure A of PPP-C-186. The glass bottles shall be packaged in tin or terne-plate slip cover closure cans conforming to PPP-C-96, type IV, coating plan B with side seams striped. The bottles shall be cushioned and the can closures taped with tape conforming to class 2 or 3 of PPP-T-60.

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5.1.1.1 Intermediate packaging. Twelve 1-pound bottles of stannic oxide, packaged as specified in 5.1.1, shall be intermediately packaged upright in a three bottle by four bottle pattern in a snug-fitting box conforming to style RSC, grade W5c of PPP-B-636. The box shall be closed in accordance with the closure requirements for class weather resistant boxes by taping as specified in the appendix to PPP-B-636.

5.1.2 Level C. Quantities of stannic oxide as specified (see 6.2) shall be packaged to provide adequate protection against contamination, deterioration, and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four boxes of stannic oxide, intermediately packaged as specified in 5.1.1.1, shall be packed for shipment in a nailed wood box conforming to class 2, style 4-1/2, for a type 2 average load of PPP-B-621. The box shall be closed and strapped using galvanized strapping in accordance with the applicable requirements specified in the appendix to PPP-B-621.

5.2.2 Level B. Stannic oxide shall be packed as specified in 5.2.1 except that the box shall conform to either a style RSC, grade V3c for a type 2 average load of PPP-B-636 or a style optional, class 2 for a type 2 average load of PPP-B-585 as specified (see 6.2). Closure shall be in accordance with the requirements for closure in the appendix to the applicable box specification.

5.2.3 Level C. Stannic oxide, packaged as specified in 5.1.2, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Department of Transportation regulations or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit and intermediate packages and shipping containers shall be marked in accordance with Department of Transportation regulations and MIL-STD-129 (see 6.2).

5.4 Palletization. When specified in the contract or order, shipping containers of 5.2 shall be palletized in accordance with the applicable requirements of MIL-STD-147 utilizing the 40- by 48-inch, double-wing pallet of MIL-P-15011 (see 6.2).

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6. NOTES

6.1 Intended use. Stannic oxide is intended for use as a general laboratory reagent.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required.
(c) Quantity required for level C packaging.
(d) Kind of box required for level B packing.
(e) Special marking, if required.
(f) Palletization, if required.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

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